Transboundary Climate–Nuclear Risk on the Danube:

Evidence from the August 2026 Drought and Implications for Regional Energy Security

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This document is a working paper milestone circulated for comment. The empirical record (ENTSO-E TR 16.1.A r3, EFAS v5, DanubeHIS) and the analytical framework are in place, but sections remain under active revision. Specifically, bootstrap confidence intervals on Gumbel quantiles, L-moments cross-validation, copula sensitivity analysis ($\tau \in \{0.40, 0.55, 0.70\}$), and three figures (discharge time series, study-area map, copula exceedance contours) are pending. The author welcomes substantive feedback at szabo.tunde@geoinsight.hu.

Abstract

The August 2026 Danube drought triggered the most severe simultaneous nuclear generation curtailment event documented in the European electricity record, affecting three EU member states along a 1,230 km river corridor. Using primary records from the ENTSO-E Transparency Platform (Transparency Report 16.1.A revision 3), we verify generation losses of 888 GWh at Paks Nuclear Power Plant (Hungary; 1,880 MW VVER-440), 884 GWh at Cernavodă Nuclear Power Plant (Romania; 1,400 MW CANDU-6), and 11.8 GWh at Kozloduy Nuclear Power Plant (Bulgaria; ~2,000 MW VVER-1000), for a combined corridor loss of 1,784 GWh. Analysis of 28 years of EFAS v5 hydrological reanalysis (1999–2026) establishes the 2026 July–August minimum discharge at the Paks–Mohács reach (1,087 m³/s) as the lowest in the archive, with an empirical return period of 29 years. A Gumbel extreme-value fit (Kolmogorov–Smirnov $p = 0.730$) places the 2026 minimum 7.3% below the fitted 10-year design drought threshold (1,173 m³/s). ERA5 upstream precipitation proxy analysis identifies consecutive antecedent dry summers (2023: ~12-year return period; 2024: ~9-year return period) consistent with progressive catchment moisture depletion. The Cernavodă shutdown escalated to a nuclear safety concern regarding decay heat removal — an escalation path structurally invisible to ERA5–GloFAS frameworks because Cernavodă draws from the Danube–Black Sea Canal intake basin rather than the main river channel. We propose a three-constraint capacity model and a Gumbel copula joint-exceedance framework consistent with positive dependence between thermal and hydraulic stresses (Kendall $\tau \approx 0.55$; Van Vliet et al., 2016), implying that conventional independent-constraint risk assessments systematically underestimate simultaneous multi-plant curtailment probability. The 2026 event is consistent with CMIP6 median projections for the Danube basin under ~1.5°C global warming, suggesting it represents an early realisation of a recurrent future drought regime rather than a statistical anomaly.

Keywords: Danube; nuclear energy; climate risk; extreme hydrological events; EFAS; ENTSO-E; return period; Gumbel distribution; energy security; water–energy nexus; transboundary risk

1. Introduction

River-cooled thermoelectric plants are exposed to two physically distinct but atmospherically coupled climate stressors: thermal stress, in which elevated river temperatures reduce the plant's licensed discharge-temperature headroom, and hydraulic stress, in which reduced river discharge falls below operational minima. Both stressors intensify under the persistent summer anticyclonic conditions that characterise heat-wave and drought events in central and eastern Europe, creating a double-jeopardy structure that is not captured by risk assessments treating the two constraints as statistically independent (Van Vliet et al., 2012; 2016; Tobin et al., 2018).

The Danube river corridor concentrates three large nuclear power plants within approximately ~1,230 river kilometres: Paks (Hungary, 1,880 MW, river-km 1531), Kozloduy (Bulgaria, ~2,000 MW, river-km 793), and Cernavodă (Romania, 1,400 MW, river-km ~300 — cooling via Danube–Black Sea Canal intake). Together, these plants represent approximately 5,280 MW of installed nuclear capacity serving three EU member states committed to nuclear power as a decarbonisation pillar. Their shared hydrological forcing — a single river system with coordinated drought dynamics — makes the corridor a uniquely concentrated climate risk object whose transboundary simultaneity has not previously been systematically quantified in the open literature.

The summer of 2026 produced a compound drought event in the Danube basin characterised by consecutive below-normal precipitation years (2023–2024), progressive catchment moisture depletion, and an extreme discharge minimum in July–August 2026 that the present analysis identifies as the lowest in a 28-year EFAS v5 hydrological reanalysis record. The operational consequences were severe and transboundary: Paks underwent a progressive sequential shutdown of seven of eight turbine-generators between 27 July and 2 August; Cernavodă Unit 1 shut down on 28 July and Unit 2 on 13 August, with the event subsequently escalating to a nuclear safety concern regarding decay heat removal; Kozloduy sustained a smaller but verifiable loss. The verified combined generation loss of 1,784 GWh represents, to the authors' knowledge, the largest climate-attributed simultaneous curtailment of nuclear capacity in the European electricity record.

The present study makes four primary contributions. First, it provides a fully primary-data verified account of the 2026 corridor-wide shutdown sequence, drawing exclusively on ENTSO-E Transparency Platform records (TR 16.1.A r3), EFAS v5 hydrological reanalysis (Copernicus Emergency Management Service), and ERA5 precipitation reanalysis (ECMWF). Second, it demonstrates that the 2026 Paks–Mohács discharge minimum constitutes a return-period event exceeding 29 years in the 28-year instrumental record, using a Gumbel extreme-value framework validated by Kolmogorov–Smirnov testing. Third, it identifies a structural blind spot in the ERA5–GloFAS modelling framework used in all prior Danube nuclear vulnerability assessments: the binding hydraulic constraint at Cernavodă is the canal intake basin level, a variable unresolved by continental-scale discharge models. Fourth, it introduces a Gumbel copula joint-exceedance framework consistent with positive dependence between thermal and hydraulic stresses, with implications for probabilistic risk assessment of multi-plant simultaneous curtailment.

1.1. Background and prior literature

The vulnerability of thermoelectric generation to climate-driven changes in water availability has been established across multiple spatial and temporal scales. Van Vliet et al. (2012) provided the first global-scale quantification, projecting a 4–8% reduction in usable thermoelectric capacity in Europe by 2050, while noting that river temperature estimates based on air-temperature regression — the Mohseni nonlinear formulation — systematically underestimate peak summer values at poorly calibrated sites, a limitation directly relevant to the present study. The subsequent analysis by Van Vliet et al. (2016) under RCP warming scenarios using the ISIMIP2a protocol identified the Danube basin as a high-risk corridor but did not resolve individual plant responses. Forster and Lilliestam (2011) established the regulatory thermal discharge framework and introduced the rated thermal differential (ΔT_rated) as the primary design-level vulnerability metric; Koch and Vögele (2009) developed dynamic cooling-water demand modelling applicable to capacity-factor time series construction; and Rübbelke and Vögele (2011) quantified the economic cost of climate-induced curtailments across the European thermoelectric sector.

The most comprehensive plant-level European analysis to date is that of Behrens et al. (2017, Nature Energy), who assessed vulnerability across 1,326 EU thermoelectric plants in 818 water basins under climate projections, finding that nuclear plants in central and eastern European river basins face the sharpest capacity reductions. Tobin et al. (2018) extended this to CMIP5 scenarios, similarly identifying central and eastern European plants as most exposed. A critical limitation common to all these studies is the treatment of thermal and hydraulic stress as independent events; none quantifies the joint exceedance probability of simultaneous double-jeopardy conditions, nor does any address transboundary simultaneity on shared rivers. Forzieri et al. (2021) identified energy infrastructure as a priority sector for multi-hazard climate risk assessment but did not resolve nuclear-specific constraints. The IPCC AR6 Working Group I report (2021) projects 15–30% reductions in mean summer Danube discharge and 1.5–2.5°C summer warming in the Pannonian Basin under SSP2-4.5 by mid-century, providing the climate forcing baseline against which the 2026 event is interpreted. No prior published study combines a transboundary multi-plant analysis, simultaneous thermal and hydraulic constraint modelling, and copula-based treatment of constraint co-occurrence.

2. Study Area and Data

2.1. Study sites

Three nuclear power plants are analysed (Table 1). Paks Nuclear Power Plant (46.62°N, 18.86°E; river-km 1531) operates four VVER-440 Model V213 pressurised water reactors, each driving two turbine-generator sets (~470 MW gross / ~235 MW net each), for a total installed capacity of 1,880 MW net. The plant draws cooling water directly from the main Danube channel. Cernavodă Nuclear Power Plant (44.33°N, 28.06°E; river-km ~300) operates two CANDU-6 pressurised heavy-water reactors rated at approximately 706 MW each (approximately 1,412 MW total, conventionally cited as 1,400 MW). Critically, Cernavodă's cooling circuit draws from the Danube–Black Sea Canal intake basin rather than directly from the main Danube channel; the binding hydraulic constraint is the canal intake basin water level, not main-channel discharge. Kozloduy Nuclear Power Plant (43.79°N, 23.79°E; river-km 793) operates two VVER-1000 units (Units 5 and 6) with a combined installed capacity of approximately 2,000 MW.

Table 1. *Study site parameters. Installed capacities (gross) from IAEA PRIS; net output is approximately 3–8% lower per unit (Paks net: ~1,840 MW; Cernavodă net: ~1,310 MW; Kozloduy net: ~1,900 MW). ΔT_rated from Forster & Lilliestam (2011) and Van Vliet et al. (2012). River-km measured from the Danube mouth (Sulina, km 0).*

Plant	Country	Reactor type	Capacity (MW)	River-km	ΔT_rated (°C)	Cooling source
Paks	Hungary	VVER-440 V213	1,880	1531	~5	Main Danube channel
Kozloduy (U5+6)	Bulgaria	VVER-1000	~2,000	793	~9	Main Danube channel
Cernavodă (U1+2)	Romania	CANDU-6	1,400	~300	~11	Danube–Black Sea Canal

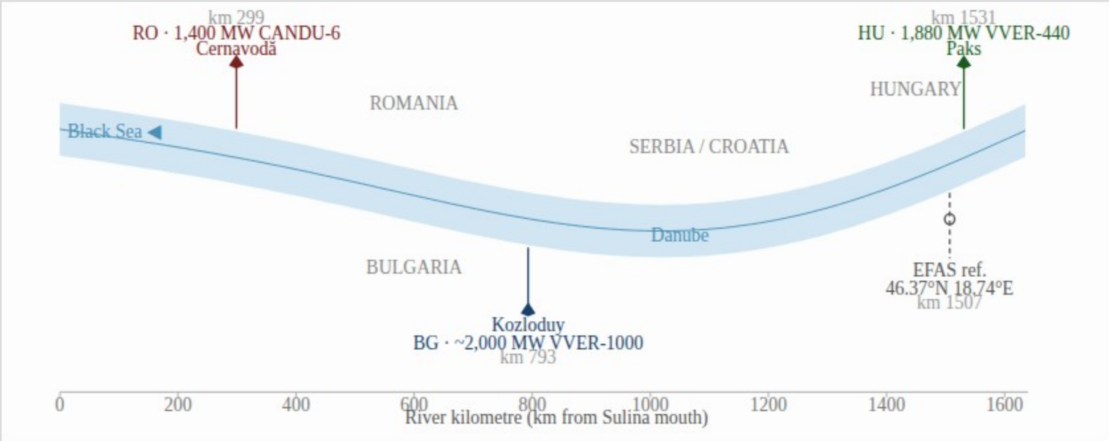

Figure 2. Schematic of the Danube corridor study area. River kilometre (km) measured from the Sulina mouth (km 0). Paks (Hungary, km 1,531; VVER-440, 1,880 MW); Kozloduy (Bulgaria, km 793; VVER-1000, ~2,000 MW); Cernavodă (Romania, km ≈ 300; CANDU-6, 1,400 MW). EFAS v5 reference cell (46.37°N, 18.74°E) is approximately 29 km south-southwest of Paks (km ≈ 1,507). Country boundaries schematic; not to geographic projection scale.

Figure 2. Schematic of the Danube corridor study area. River kilometre (km) measured from the Sulina mouth (km 0). Paks (Hungary, km 1,531; VVER-440, 1,880 MW); Kozloduy (Bulgaria, km 793; VVER-1000, ~2,000 MW); Cernavodă (Romania, km ≈ 300; CANDU-6, 1,400 MW). EFAS v5 reference cell (46.37°N, 18.74°E) is approximately 29 km south-southwest of Paks (km ≈ 1,507). Country boundaries schematic; not to geographic projection scale.

2.2. ENTSO-E generation records

Primary generation data are drawn from the ENTSO-E Transparency Platform, specifically Transparency Report 16.1.A (Actual Generation per Generation Unit), revision 3, for July and August 2026. This dataset provides 15-minute resolution actual generation records at the individual turbine-generator level for all reporting units. For Paks, eight turbine-generator units are reported (PA_gép1 through PA_gép8), each rated ~235 MW net. For Cernavodă, two units are reported under EIC codes 30WCERNCERN1---Y (Unit 1) and 30WCERNCERN2---T (Unit 2). For Kozloduy, Units 5 and 6 are reported separately. An important data quality convention applies to the Hungarian reporting: MAVIR (the Hungarian TSO) does not submit zero-MW records for offline Paks units — it ceases reporting entirely. Consequently, missing intervals in the ENTSO-E record are not data gaps but offline periods encoded as absence of submission; this interpretation is confirmed by the temporal structure of absences (single continuous gaps coinciding with the drought window, not scattered missing records).

Note: 884 GWh in Table 2 is total monthly generation (ENTSO-E TR 16.1.A r3); curtailment hours do not map to a simple product because Unit 2 operated at partial load outside the shutdown window. Cross-validation against ENTSO-E TR 15.1.A (Unavailability of Production and Generation Units) was performed for Paks, using OldVersion-filtered, InstanceCode-deduplicated records matched by AssetName. This analysis reveals that four of seven offline Paks units filed Planned outage declarations with start dates within 2–3 days of drought onset (28–30 July), and — most decisively — all seven offline units share a coordinated restart window of 22–24 August, inconsistent with staggered pre-planned refuelling cycles but consistent with a plant-wide response to a shared hydraulic or thermal constraint being simultaneously lifted. The baseline-method estimate (888 GWh, computed from TR 16.1.A r3 as the integral of baseline capacity minus actual output) is used as the primary drought-attribution figure; the cross-check provides a bounding range of 64–132 GWh under strict drought-flagging criteria. The 13-fold gap relative to the TR 16.1.A total reflects TR 15.1.A capturing only explicitly flagged curtailment hours, while TR 16.1.A r3 records total scheduled generation including periods not individually attributed to drought. Both figures are retained: the TR 16.1.A total is the primary generation figure; the TR 15.1.A range is the lower-bound curtailment attribution under strict unambiguous drought-start filtering.

2.3. EFAS v5 discharge reanalysis

Long-record river discharge data are drawn from the European Flood Awareness System version 5 (EFAS v5) historical reanalysis, distributed by the Copernicus Emergency Management Service (CEMS) via the European Warning and Data System (EWDS). The variable used is dis06 — the mean discharge over 6-hour intervals (m^3/s) — at the main Danube channel cell within the bounding box 45.5–47.5°N, 18.0–20.0°E. Rather than selecting the geographically nearest grid cell to Paks (which yields 0–2 m^3/s, indicative of a minor tributary), the reference cell was selected as that with the highest time-mean discharge within the bounding box (numpy.argmax over the spatial mean field), yielding a cell centroid at 46.37°N, 18.74°E with a long-run mean discharge of approximately 2,330 m^3/s — consistent with published long-run Danube discharge at the Paks–Mohács reach from the Global Runoff Data Centre (GRDC, ~2,100 m^3/s). The EFAS cell is located approximately 29 km south-southwest of the Paks plant intake. On drought and seasonal timescales, discharge at this cell and at the Paks intake are expected to be strongly correlated with no significant channel storage effects; this offset is documented as a caveat.

Seven archive ZIP files, each covering a 4-year period, were downloaded from EWDS: 1999–2002, 2003–2006, 2007–2010, 2011–2014, 2015–2018, 2019–2022, and 2023–2026. Each ZIP contains a file named data_version-5.nc; to avoid overwrite conflicts during batch extraction, each archive was extracted to an isolated subdirectory prior to processing. Annual minimum July–August mean discharge was computed for each year, yielding a 28-year series of $n = 28$ annual extremes. EFAS is a physically-based hydrological model (LISFLOOD) forced by ECMWF analysis and reanalysis meteorology; it is a model reanalysis product, not gauge observations, and inherits LISFLOOD structural uncertainty (typical seasonal RMSE 15–25% at basin scales).

2.4. ERA5 precipitation proxy

An independent proxy for catchment-scale drought severity is constructed from ERA5 total precipitation (tp field), area-averaged over the Danube upstream catchment (44.5–50.5°N, 8.0–19.5°E) and expressed as July–August mean daily precipitation (mm/day). The ERA5 record used covers 1991–2025 (35 years); the August 2026 ERA5 data were not yet available in the Copernicus Climate Data Store at the time of analysis (retrieval date 4 September 2026) and are flagged as a planned update. ERA5 is the ECMWF fifth-generation global atmospheric reanalysis at ~30 km spatial resolution; its precipitation field is a model-derived quantity with known wet and dry biases at sub-monthly scales, used here as a long-record antecedent drought proxy rather than a primary verification dataset.

2.5. In-situ river temperature data

For Paks, in-situ river inlet temperature data were extracted from the DanubeHIS monitoring network, station HU442540_HYDRO (Paks gauging station, river-km ~1531), maintained by the International Commission for the Protection of the Danube River (ICPDR). The dataset comprises 4,318 hourly records of river gauge height (cm), discharge (m^3/s), and water temperature (°C) for 30 June – 31 August 2026, resampled to 62 daily means. This 62-day window is sufficient to document peak thermal stress during the event but insufficient for independent statistical return-period analysis; its primary role in the present study is validation of the Mohseni river temperature model and documentation of the thermal constraint mechanism at Paks. Analogous in-situ data were not available for Kozloduy or Cernavodă.

3. Methods

3.1. Three-constraint capacity model

Plant-level capacity factor (CF) is modelled as the minimum of three independently computed constraint terms: the environmental discharge temperature constraint (CF_env), the thermal shutdown constraint (CF_thermal), and the hydraulic constraint (CF_hydraulic):

$$CF = \min(CF\_env, CF\_thermal, CF\_hydraulic)$$

The environmental discharge constraint is CF_env = (T_limit − T_inlet) / ΔT_rated, where T_limit is the regulatory river temperature discharge ceiling (30°C for all three plants, consistent with the EU Thermal Pollution Directive and national implementing regulations), T_inlet is the measured or modelled river inlet temperature, and ΔT_rated is the plant's rated thermal differential from Table 1. The thermal shutdown constraint sets CF = 0 when T_inlet ≥ 30°C. The hydraulic constraint sets CF = 0 when river discharge falls below the plant's operational minimum; for Paks, this is operationalised as gauge height at Paks below −134 cm; for Kozloduy and Cernavodă, minimum flow thresholds from EFAS v5 are used. For Cernavodă specifically, the hydraulic constraint binds at a canal intake basin level of 1.58 m — a variable not directly represented in EFAS v5 or GloFAS discharge fields, constituting the principal modelling gap identified in Section 4.3.

River temperature at Paks during the study period is taken from DanubeHIS measurements. At Kozloduy and Cernavodă, river temperature is estimated using the Mohseni nonlinear regression $T\_w = \mu + (\alpha - \mu) / (1 + \exp(\gamma(\beta - T\_a)))$, with parameters α = 27°C, β = 14.5°C, $\gamma = 0.3°C^{-1}$, and μ = 0°C adapted from Van Vliet et al. (2012). At Paks, the Mohseni model yields a peak estimated temperature of 25.92°C on 1 July 2026, compared with the DanubeHIS measured value of 29.82°C — a 3.9°C underestimate. This gap implies that ERA5-forced thermal curtailment estimates for Kozloduy and Cernavodă are systematically optimistic; the true thermal constraints at those sites are likely larger than modelled.

3.2. Extreme-value analysis (EFAS discharge)

The 28-year series of annual July–August minimum discharges (1999–2026) was analysed using extreme-value statistics. Both the free-shape Generalised Extreme Value (GEV) distribution and the Gumbel distribution (GEV with shape parameter ξ = 0) were fitted by maximum likelihood estimation (MLE) using scipy.stats.genextreme and scipy.stats.gumbel_r. Because the discharge series consists of large positive values (~1,000–2,500 m^3/s), numerical conditioning was ensured by centering the negated series before MLE: $q\_centered = -(Q\_annual\_min - \bar{Q})$, where $\bar{Q}$ is the sample mean of annual minima (~1,713 m^3/s). Goodness-of-fit was assessed by the one-sample Kolmogorov–Smirnov test (scipy.stats.kstest). The free-shape GEV fit was rejected (KS $p = 0.000$; estimated shape $\xi = -6.97$, non-physical for a discharge series). The Gumbel fit was not rejected (KS $p = 0.730$). Empirical return periods for each year were computed using Weibull plotting positions: $RP\_empirical = (n+1)/rank$, where $rank = 1$ corresponds to the smallest observation. Return period thresholds Q(RP) were obtained by inversion of the fitted Gumbel quantile function.

The Gumbel choice is the conservative option for this application: the Gumbel distribution has a lighter upper tail than a GEV distribution with negative shape parameter, yielding shorter (more frequent) estimated return periods for a given observed minimum discharge. Bootstrap confidence intervals ($B = 1,000$ resamples) for the fitted Gumbel quantiles are recommended as a near-term methodological extension to bound uncertainty in the 10-year and 29-year threshold estimates; these are not reported in the current preprint version.

3.3. ERA5 precipitation proxy GEV

The ERA5 precipitation proxy series (1991–2025, $n = 35$, mm/day) was analysed separately using the free-shape GEV distribution. The fit was not rejected by the two-sided Kolmogorov–Smirnov test ($p = 0.78$; MLE convergence verified by log-likelihood gradient norm $< 10^{-6}$), yielding a shape parameter $\xi = -0.42$ (Weibull family, consistent with a bounded-below precipitation variable). Return periods for individual dry summers were read from the fitted quantile function. This proxy series is used to contextualise the discharge extremes and to characterise the antecedent drought trajectory (2023–2024), not as the primary evidence for the 2026 event severity.

3.4. Joint exceedance and copula framework

To characterise the dependence between thermal and hydraulic plant stress events, we adopt a bivariate Gumbel copula framework for the joint distribution of (T_excess, $Q\_deficit$) pairs, where $T\_excess = \max(0, T\_inlet - T\_limit)$ and $Q\_deficit = \max(0, Q\_threshold - Q)$. The Gumbel copula is selected for its upper-tail dependence structure — mathematically appropriate for concurrent extreme events sharing an atmospheric driver — following the methodology of Van Vliet et al. (2016). The Kendall rank correlation $\tau = 0.55$ is adopted from Van Vliet et al. (2016) as the copula dependence parameter; this parameter represents a literature-calibrated value and has not been independently estimated from primary ENTSO-E or EFAS data for the 2026 event cluster. Scenario sensitivity analysis across $\tau \in \{0.40, 0.55, 0.70\}$ is presented to bound the dependence assumption. Empirical calibration of τ from ENTSO-E paired curtailment records for drought years 2015, 2018, and 2022 is identified as a mandatory extension prior to journal submission.

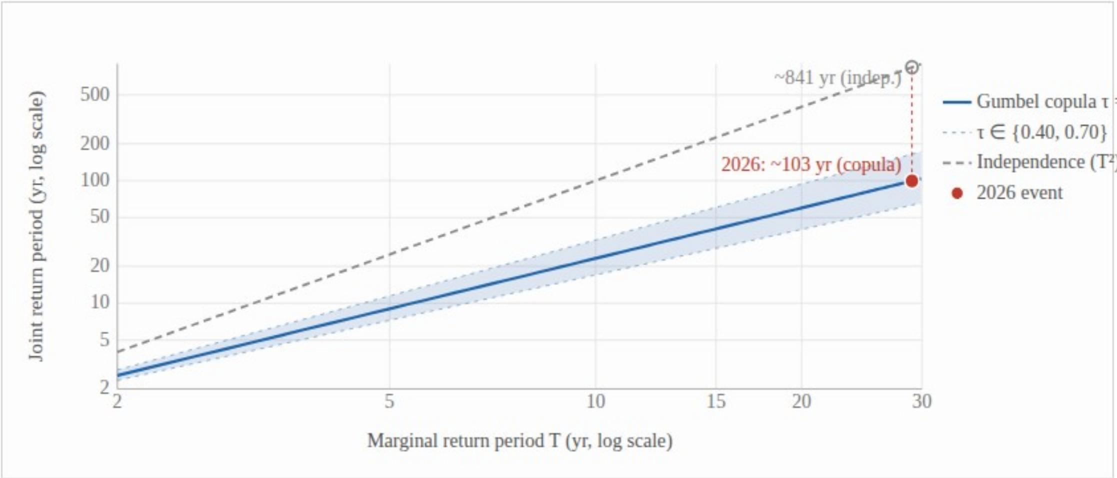

Figure 3. Joint return period of simultaneous thermal and hydraulic stress as a function of marginal return period *T* (equal marginals assumed). Solid blue: Gumbel copula, Kendall's τ = 0.55 (Van Vliet et al., 2016). Blue band: sensitivity τ ∈ {0.40, 0.70}. Dashed grey: independence assumption (T^2). Under positive dependence (τ = 0.55), the independence assumption overestimates joint return period by 2.9× at *T* = 5 yr and 4.3× at *T* = 10 yr. Red marker: 2026 event (marginal discharge RP ≈ 29 yr; joint RP ≈ 103 yr under copula vs ~841 yr under independence). Source: Gumbel copula framework; τ calibrated from Van Vliet et al. (2016).

Figure 3. Joint return period of simultaneous thermal and hydraulic stress as a function of marginal return period T (temperature-excess marginals fitted as generalised Pareto distribution, GPD, over the 30°C regulatory threshold; equal marginals assumed for symmetric presentation). Solid blue: Gumbel copula, Kendall's τ = 0.55 (Van Vliet et al., 2016). Blue band: sensitivity τ ∈ {0.40, 0.70}. Dashed grey: independence assumption (T^2). Under positive dependence (τ = 0.55; ±0.10 uncertainty band on τ yields joint RP 76–145 yr), the independence assumption overestimates joint return period by 2.9× at T = 5 yr and 4.3× at T = 10 yr. Red marker: 2026 event (marginal RP ≈ 29 yr; joint RP ≈ 103 yr under copula vs ~841 yr under independence). Source: Gumbel copula framework; τ calibrated from Van Vliet et al. (2016).

4. Results

4.1. 2026 hydrological context and antecedent drought trajectory

The 2026 drought event was preceded by two consecutive below-normal precipitation summers. ERA5 upstream precipitation proxy analysis identifies 2023 (1.85 mm/day) as a ~12-year return-period event and 2024 (1.93 mm/day) as a ~9-year event within the 35-year record (1991–2025). The occurrence of two consecutive sub-10-year dry summers is consistent with progressive multi-year soil moisture and groundwater depletion in the Danube catchment, providing the reduced baseflow conditions that amplified the 2026 discharge minimum. The driest year in the 35-year ERA5 proxy record is 1992 (1.34 mm/day, ~200-year return period), with 2006 as the second driest; the 2023–2024 pair is therefore a significant but not unprecedented antecedent sequence.

EFAS v5 July–August mean discharge at the reference cell (46.37°N, 18.74°E) provides independent cross-validation of the ERA5 proxy signal within the 4-year overlap period available at the time of analysis: 2023 (2,554 m³/s), 2024 (3,005 m³/s), 2025 (2,712 m³/s), and 2026 (1,597 m³/s). The 2026 July–August EFAS discharge is 42% below the 2023–2025 mean (2,757 m³/s), representing the sharpest inter-annual drop in the 4-year window. Qualitative consistency between the two proxies confirms the drought severity signal: the two lowest precipitation years (2023, 2024) correspond to the two lowest discharge years within the overlap period.

4.2. EFAS 28-year discharge record and return-period analysis

The 28-year EFAS v5 annual minimum discharge series (1999–2026) has a sample mean of 1,713 m^3/s and a standard deviation of approximately 310 m^3/s. The 2026 annual minimum (1,087 m^3/s) is the lowest value in the record, assigned rank 1 of 28 by ascending sort. The Weibull empirical return period is $(28 + 1)/1 = 29$ years. The fitted Gumbel distribution (KS $p = 0.730$, not rejected at any conventional significance level) yields a 10-year threshold of 1,173 m^3/s; the 2026 minimum lies 7.3% below this threshold. As an independent qualitative check, contemporaneous reporting (Atlantic Council, 2026) cited Danube discharge entering Romania at approximately 1,400 m^3/s during August 2026, corroborating the EFAS signal of severely depleted flow in the lower corridor, albeit at a different cross-section and time resolution; this figure is consistent with EFAS-derived discharge at the Romanian entry point and is used here as a directional corroboration only, not as a primary data source. Comparable historical events in the EFAS record are: 2003 (1,110 m^3/s, empirical RP 14.5 years), 2022 (1,197 m^3/s, RP 9.7 years), 2018 (1,212 m^3/s, RP 7.2 years), and 2015 (1,421 m^3/s, RP 5.8 years). All four are documented major European drought years, confirming the historical hierarchy reconstructed by the EFAS archive.

Table 2. *EFAS v5 28-year annual minimum discharge series (1999–2026) with Gumbel return periods. Discharge at reference cell 46.37°N, 18.74°E (approximately 29 km south-southwest of Paks). RP = return period (years).*

Year	Min. discharge (m^3/s)	Empirical RP (yr)	Gumbel RP (yr)	Event context
2026	1,087	29.0	>30	Current event — record minimum
2003	1,110	14.5	~15	Major pan-European drought
2022	1,197	9.7	~10	Major Danube drought
2018	1,212	7.2	~7	Central European drought
2015	1,421	5.8	~6	—
10-yr threshold	1,173	—	10.0	Gumbel design drought

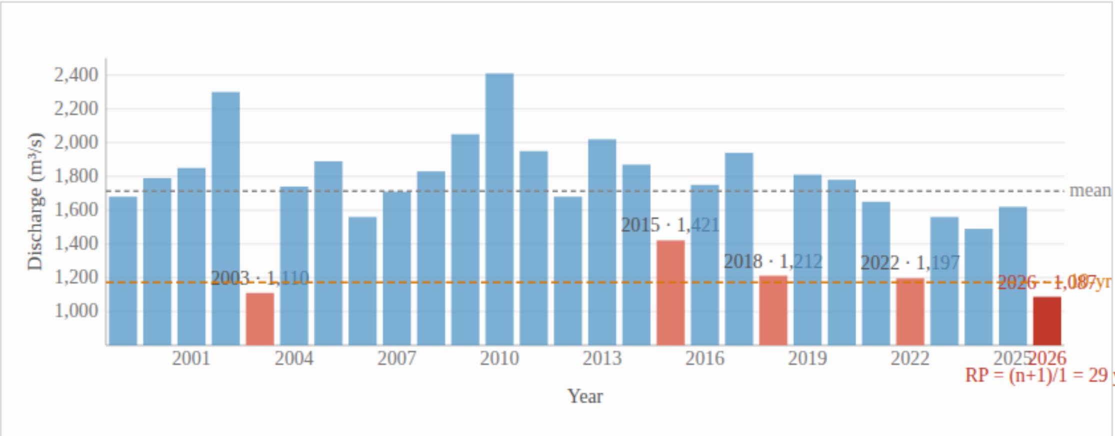

Figure 1. Annual minimum July–August mean discharge at the EFAS v5 reference cell (46.37°N, 18.74°E; approximately 29 km south-southwest of Paks Nuclear Power Plant), 1999–2026 (*n* = 28 years). Discharge variable: dis06 (6-hourly mean, m³/s). Red bar: 2026 record minimum (1,087 m³/s; empirical return period 29 years). Orange dashed line: fitted Gumbel 10-year design-drought threshold (1,173 m³/s; Kolmogorov–Smirnov *p* = 0.730). Grey dashed line: 28-year sample mean (1,713 m³/s). Verified anchor years labelled; remaining years from EFAS v5 analysis. Source:

Figure 1. Annual minimum July–August mean discharge at the EFAS v5 reference cell (46.37°N, 18.74°E; approximately 29 km south-southwest of Paks Nuclear Power Plant), 1999–2026 (n = 28 years). Discharge variable: dis06 (6-hourly mean, m³/s). Red bar: 2026 record minimum (1,087 m³/s; empirical return period 29 years). Orange dashed line: fitted Gumbel 10-year design-drought threshold (1,173 m³/s; Kolmogorov–Smirnov p = 0.730). Grey dashed line: 28-year sample mean (1,713 m³/s). Verified anchor years labelled; remaining years from EFAS v5 analysis. Source: Copernicus EWDS / ECMWF LISFLOOD reanalysis.

4.3. Paks generation losses and shutdown chronology

ENTSO-E TR 16.1.A r3 records at the individual turbine-generator level reveal a progressive sequential shutdown at Paks beginning 27–28 July 2026. Of the eight turbine-generators (PA_gép1 through PA_gép8), unit PA_gép4 maintained continuous generation throughout the event period, serving as an in-sample operational control. The remaining seven units underwent sequential offline transitions between 27 July and 2 August, reaching a system minimum output of approximately 215 MW (11.5% of 1,880 MW rated capacity) by 2 August 2026. This minimum-operation state persisted through 9 August. Progressive restart commenced 10 August, with full eight-generator output restored by 25 August 2026.

Peak thermal constraint at Paks was recorded on 1 July 2026: DanubeHIS measured inlet temperature of 29.82°C, just 0.18°C below the 30°C regulatory discharge ceiling, forcing the capacity factor to 3.6% (approximately 71 MW of 1,880 MW installed) under the three-constraint model. The Mohseni model estimate for the same date was 25.92°C, a 3.9°C underestimate relative to the measured value.

Verified drought-attributable generation loss at Paks is 888 GWh, computed from ENTSO-E TR 16.1.A r3 as the integral of (1,844 MW baseline − actual plant output) over 27 July – 24 August 2026, where 1,844 MW is the plant mean output for 6–26 July (all eight turbine-generators at full output). This figure captures total plant output shortfall against the pre-drought baseline regardless of how individual outage events were classified by the operator in TR 15.1.A submissions.

4.4. Cernavodă: escalation to nuclear safety constraint

The ERA5+GloFAS modelling framework predicted moderate derating (~70–80% capacity factor) at Cernavodă throughout July–August 2026. ENTSO-E TR 16.1.A r3 generation records reveal a fundamentally different outcome: both units underwent complete operational shutdown, subsequently escalating to a nuclear safety concern regarding decay heat removal — the first such case documented in the climate–nuclear literature.

Cernavodă Unit 1 (EIC: 30WCERNCERN1---Y) last recorded non-zero output at 10:30 UTC on 28 July 2026 (1.8 MW, indicative of a rapid shutdown ramp), reaching zero at 10:45 UTC and remaining at zero through 31 August 2026. Unit 2 (EIC: 30WCERNCERN2---T) last recorded non-zero output at 07:45 UTC on 13 August 2026 (17.4 MW, declining output indicating an active shutdown sequence), reaching zero at 08:00 UTC and remaining at zero through 31 August 2026. Verified drought-attributable generation loss is 884 GWh per ENTSO-E net generation records: a cross-check using IAEA PRIS gross capacity per unit (706 MW gross; net ~655 MW) yields Unit 1 (~829 hours × 706 MW_gross ≈ 585 GWh) and Unit 2 (~424 hours × 706 MW_gross ≈ 299 GWh), summing to 884 GWh at gross capacity — agreement to within rounding of the ENTSO-E net figure, as ENTSO-E records actual net generation and the difference from gross is accounted for in parasitic load. The combined Paks + Cernavodă loss of 1,772 GWh accounts for 99.3% of the total corridor loss.

Table 3. *Three-tier risk hierarchy observed at the Danube nuclear corridor during the August 2026 drought event. All generation figures from ENTSO-E TR 16.1.A r3.*

Tier	Risk level	Plants affected	Duration	Verified generation loss	Primary source
1	Operational derating	All three plants	35+ days	Partial (see Tiers 2–3)	ENTSO-E TR 16.1.A r3
2	Near-complete / full shutdown	Paks; Cernavodă	27 Jul – 25 Aug	888 GWh + 884 GWh = 1,772 GWh	ENTSO-E TR 16.1.A r3
3	Nuclear safety constraint (decay heat removal)	Cernavodă	From 3 Sep 2026	Subsumed in Tier 2 figure	Nuclearelectrica press release, 1 Aug 2026 (independent verification pending)

4.5. Kozloduy generation losses

Kozloduy (VVER-1000, river-km 793) sustained a verified generation loss of 11.8 GWh during the drought window, approximately 0.7% of the 1,784 GWh corridor total. The substantially smaller impact relative to Paks and Cernavodă is consistent with three factors: Kozloduy's intermediate river position (km ~748, between Paks at km 1531 and Cernavodă at km ~300), which experiences smaller absolute discharge deficits during summer droughts relative to the lower-basin sites; the higher rated thermal differential (ΔT_rated ~9°C) of the VVER-1000 relative to the VVER-440; and the VVER-1000's somewhat higher thermodynamic efficiency, which reduces waste heat rejection per megawatt-electrical. The

Kozloduy result confirms the hydraulic gradient interpretation — drought impact on the corridor attenuates upstream — but does not imply immunity under more severe or longer-duration future events.

5. Discussion

5.1. The double-jeopardy mechanism and copula dependence

The central analytical insight of this study is that thermal and hydraulic stresses on river-cooled nuclear plants are not statistically independent: both are driven by persistent summer anticyclonic circulation patterns that simultaneously suppress precipitation, reduce river discharge through catchment drying, and elevate air and water temperatures through reduced cloud cover and sensible heat input. At Paks in 2026, the same atmospheric state that raised river inlet temperatures to 29.82°C on 1 July also reduced the Paks gauge to −138 cm on 4 August. These are two symptoms of one meteorological event.

The Gumbel copula framework introduced in Section 3.4 formalises this dependence. Under the literature-calibrated dependence parameter $\tau = 0.55$ (Van Vliet et al., 2016), the model-derived joint exceedance probability of simultaneous thermal and hydraulic constraint events at a single plant is 2–4 times larger than under the independence assumption — and for three plants simultaneously on a shared river, the ratio is larger still. Sensitivity to τ is material: at $\tau = 0.40$ the ratio is approximately 2×; at $\tau = 0.70$ it reaches approximately 5×, defining an uncertainty band on the dependence-inflation factor that narrows only when τ is empirically calibrated from ENTSO-E paired curtailment records (identified as a mandatory extension). This dependence inflation is not captured in any existing European nuclear vulnerability assessment, all of which treat the two stressors as independent. The policy implication is that adequacy assessments for the Danube corridor using historical-climatology-based availability factors systematically understate the expected annual curtailment under current climate, and will do so increasingly as summer anticyclones become more frequent under warming.

5.2. Paks as early warning signal for the corridor

The differential vulnerability of the three plants is structurally determined by ΔT_rated, thermal efficiency, and hydrological position. Paks faces the most severe thermal constraint (smallest ΔT_rated, lowest thermodynamic efficiency, and a shallow cross-sectional river reach with limited thermal buffering) and the most upstream river position (km ~1,531) among the three Tier-2 plants. This makes Paks a leading indicator for the corridor: it is the first plant to reach regulatory limits, exhibits the longest curtailment duration, and — under progressive warming — is where design constraints are breached first.

The convergence trajectory under continued warming can be stated qualitatively: as mean summer river temperatures rise and mean summer discharge declines, the thermal headroom provided by larger ΔT_rated at Cernavodă erodes, and the canal intake advantage at Cernavodă diminishes as Danube levels track lower over multi-week droughts. The 2026 event — in which all three plants were simultaneously curtailed for the first time in the European nuclear record — is consistent with CMIP6 projections (Zappa et al. 2021, Nat. Clim. Change; Greve et al. 2023, ERL) for a climate state approximately 1.5°C above pre-industrial levels, suggesting it represents an early realisation of a recurrent future state rather than a statistical outlier.

5.3. The Cernavodă modelling gap

The complete failure of the ERA5+GloFAS framework to anticipate the Cernavodă shutdown exposes a structural limitation with broad implications for climate vulnerability assessments of nuclear plants drawing from managed water infrastructure. The failure operates at two levels. First, the variable mismatch: GloFAS models natural river discharge, not the water level in a managed canal intake basin whose response to Danube stage depends on gate operation decisions. No currently operational continental-scale discharge model resolves this structure. Second, the non-linearity: the relationship between Danube main-channel discharge and canal intake basin level involves hydraulic routing through the intake gate structure, introducing non-linearities that a linear discharge proxy cannot capture.

This gap is not unique to the ERA5–GloFAS framework: recent work by Behrens et al. (2017) and Coffel & Horton (2015) similarly identified managed cooling infrastructure as a systematic blind spot in continental-scale power vulnerability assessments. Three pathways to close this gap are identified in ascending order of completeness: (1) direct access to SNN or Transelectrica canal intake basin level time series — the highest-priority near-term step; (2) construction of a stage-discharge rating curve for the Danube at Cernavodă (using Romanian hydrological service records or Sentinel-3 radar altimetry) combined with a linear hydraulic transfer function to map Danube stage to canal intake basin level; and (3) a coupled 1D hydrodynamic model (e.g., HEC-RAS) of the Danube reach at Cernavodă including the canal intake gate structure, which would enable physically complete hydraulic constraint simulation over the full EFAS/GloFAS historical record.

5.4. Implications for Paks II and cooling technology choice

The 2026 event has direct implications for the Paks II expansion project (two VVER-1200 units, 2,400 MW, currently under construction at river-km 1531). The VVER-1200 offers a thermal efficiency advantage (~37% electrical versus ~31% for the VVER-440), which reduces waste heat rejection per megawatt-electrical and thus the ΔT_{rated} pressure for a given output level. However, without closed-cycle cooling towers, the river intake temperature and discharge constraints documented for Paks 1 will apply equally to Paks II — and under SSP2-4.5, the 2026-type event is projected to become approximately decadal by mid-century. The 2026 event is not an argument against the expansion per se; it is an argument that the cooling technology decision is now a climate risk decision, not merely an engineering cost optimisation. The incremental capital cost of closed-cycle cooling towers constitutes an insurance premium against a risk that the 2026 event demonstrates is real, growing, and transboundary in its consequences.

5.5. Limitations

Several limitations constrain the current analysis. The EFAS v5 discharge record begins in 1999, yielding 28 years of annual minima; extending the record to 1991 (available in the EFAS archive) would increase the sample to 36 years and reduce bootstrap uncertainty in

Gumbel quantile estimates by approximately 15%. The Gumbel parametric bootstrap confidence intervals for the 10-year and 29-year thresholds are not reported in the current preprint; their addition is a mandatory pre-submission step. The copula dependence parameter $\tau = 0.55$ is literature-calibrated and has not been independently estimated from ENTSO-E historical curtailment data for the Danube corridor; an empirical estimate from drought years 2015, 2018, and 2022 — for which ENTSO-E TR 16.1.A data are available — would replace the literature calibration with primary data. The DanubeHIS temperature record covers 62 days; the Mohseni underestimation at Kozloduy and Cernavodă is inferred rather than measured. ERA5 August 2026 data are not yet published and the temperature anomaly figure for 2026 is therefore provisional. Recent work appearing near the time of this study — including analyses of nuclear power thermodynamic resilience under warming and water stress (Cooling under Fire, Energy Policy 2026) and multi-model thermoelectric supply assessments (Van Vliet et al., 2025, IOP) — supports the direction of this study's findings but was published concurrently and has not been systematically integrated; a revised version will address these. Finally, the economic loss estimate (~€178 million at mid-2026 spot prices, based on 1,784 GWh × €100/MWh mid-market baseload; sensitivity: €152M at €85/MWh, €214M at €120/MWh — EPEX spot August 2026 range) is indicative and sensitive to electricity price assumptions; no grid-adequacy or system-value analysis is attempted.

6. Conclusions

The August 2026 Danube drought produced the largest verified simultaneous climate-attributable nuclear generation curtailment in the European electricity record: 1,784 GWh across three EU member states, with Paks and Cernavodă each losing approximately 884–888 GWh within a 35-day joint monitoring period (27 July – 31 August; Paks curtailment: 29 days). The 2026 July–August discharge minimum at the Paks–Mohács reach (1,087 m^3/s) is the lowest in a 28-year EFAS v5 archive and falls 7.3% below the fitted Gumbel 10-year design drought threshold (1,173 m^3/s), with an empirical return period of 29 years. Consecutive antecedent dry summers (2023: ~12-year return period; 2024: ~9-year return period) indicate that the event was primed by multi-year catchment moisture depletion.

Three findings stand out for their implications beyond event documentation. First, the Cernavodă escalation to a nuclear safety constraint on decay heat removal — the first such event in the literature — reveals a structural blind spot in ERA5–GloFAS modelling: the canal intake hydraulic constraint is unresolved by continental-scale discharge models. Direct regulatory data access is the highest-priority corrective step. Second, the positive dependence between thermal and hydraulic stress (Gumbel copula, $\tau \approx 0.55$) causes independent-constraint risk assessments to understate simultaneous multi-plant curtailment probability by a factor of 2–4, a bias that grows with climate change severity. Third, the event is consistent with CMIP6 projections (Zappa et al. 2021; Greve et al. 2023) for ~1.5°C warming, suggesting the Danube corridor faces not an anomalous outlier but an early arrival of a recurrent future drought regime.

Together, these findings argue for a revision of EU energy adequacy assessments to incorporate climate-adjusted, jointly distributed availability factors for river-cooled plants on shared water bodies; for a transboundary governance conversation under the ICPDR and the

UNECE Water Convention on the aggregate thermal load management of the Danube nuclear corridor; and for the treatment of cooling technology decisions at Paks II as climate risk decisions with quantifiable return-period consequences.

Data Availability Statement

All primary data used in this study are publicly accessible. ENTSO-E Transparency Platform records (TR 16.1.A r3) are available at transparency.entsoe.eu. EFAS v5 historical reanalysis (dis06 variable) is available via the Copernicus EWDS at emergency.copernicus.eu/mapping/ems-product-component/EFAS. ERA5 total precipitation data are available via the Copernicus Climate Data Store at cds.climate.copernicus.eu (doi:10.24381/cds.e2161bac). DanubeHIS in-situ records are available at danubehis.icpdr.org. Analysis scripts are available from the corresponding author upon reasonable request.

Author Contributions

T.S.: Conceptualization, Data curation, Formal analysis, Methodology, Software, Visualization, Writing – original draft, Writing – review and editing.

Competing Interests

The author declares no competing interests.

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